MACHINE LEARNING PROJECT

Smart Customer Churn & *Retention* Intelligence System

An end-to-end machine learning workflow that predicts which customers are likely to leave, ranks them by revenue at risk and recommends a retention action for each.

12,000	0.855	1,879	3
CUSTOMERS SCORED	TEST ROC-AUC	HIGH-RISK FLAGGED	MODELS COMPARED

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1 · OVERVIEW

The problem and *the approach*

Keeping an existing customer is usually cheaper than winning a new one, but a retention team has limited time. It needs to know which customers are most likely to leave, how much revenue is at stake, and what to do for each one.

This project builds that workflow end to end. It generates a 12,000-customer telecom-style dataset, compares three machine learning models, selects one on validation data, tests it once on held-out customers, and turns the scores into risk bands and recommended actions.

Data → 70 / 15 / 15 split → 3 models → Threshold → Risk bands → Actions

RESULTS AT A GLANCE

0.855	0.684	81.2%	62.6%
ROC-AUC, test set	PR-AUC (baseline 0.234)	Accuracy, test set	Recall on churners

WHAT THE SYSTEM PRODUCES

- A churn probability for every customer, exported to CSV for Power BI.
- A High, Medium or Low risk band, using cut-offs at 25% and 50% probability.
- A recommended retention action, chosen by risk band and by ticket or payment history.
- An annual charge exposure figure (monthly charges × 12 × churn probability) used to rank who to contact first.

> **Data disclosure.** The dataset is simulated, not real customer records. Churn outcomes are sampled from a probability model with built-in relationships. The scores therefore demonstrate the workflow and must not be read as performance on real customers.

2 · DATA AND METHOD

What goes *into* the model

Each of the 12,000 customers has 11 input features. Overall, 23.5% of customers churn, so churners are the minority class and accuracy alone can hide missed churners.

Feature	Type	Meaning
tenure_months	Numeric	Months as a customer (1 to 72)
monthly_charges	Numeric	Monthly bill, 20 to 125
contract_type	Category	Month-to-month, One year, Two year
support_tickets, late_payments	Count	Service and payment friction
customer_satisfaction	Numeric	Score from 1 to 10
service_downtime_hours	Numeric	Hours of service downtime
monthly_usage_hours, usage_decline_flag	Numeric / flag	Engagement level and whether usage fell
autopay_enabled, active_addon_count	Flag / count	Stickiness signals

SPLITTING AND PREPROCESSING

Set	Customers	Churn rate	Used for
Training (70%)	8,400	23.49%	Fitting the models
Validation (15%)	1,800	23.50%	Model and threshold selection
Test (15%)	1,800	23.44%	One final, honest evaluation

The split is stratified so each set keeps the same churn rate. Preprocessing is fitted on training data only, inside pipelines, to avoid leakage: median imputation and scaling for numbers, one-hot encoding for categories. CatBoost reads the categorical column directly.

MODEL SELECTION

Logistic Regression, Random Forest and CatBoost were compared on validation data. CatBoost was selected on the highest PR-AUC, a measure suited to a minority class.

Model	Accuracy (0.50)	ROC-AUC	PR-AUC
CatBoost (selected)	**0.8311**	**0.8685**	**0.6731**
Logistic Regression	0.8311	0.8655	0.6679
Random Forest	0.8106	0.8545	0.6408

The three models are close. That is expected on data generated from a simple probability model, and it is a reminder that model choice matters less here than the threshold and the actions built on top.

3 · EVALUATION

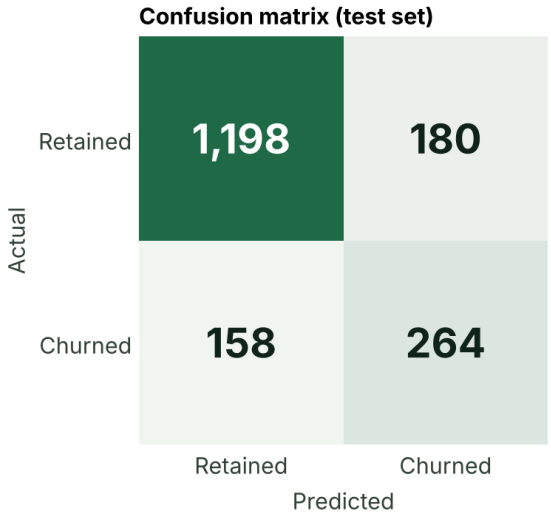
How well does it *perform?*

DECISION THRESHOLD

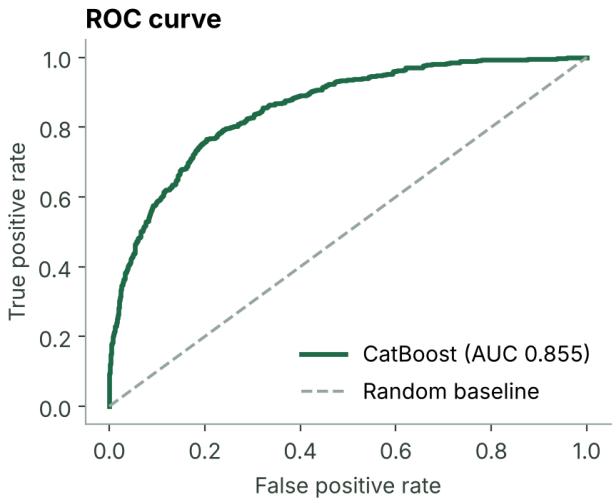
A default 0.50 cut-off misses too many churners. The threshold was chosen on validation data only: the highest accuracy among thresholds that still catch at least 65% of churners. This gave **0.355**, with validation accuracy 0.824, precision 0.617 and recall 0.662.

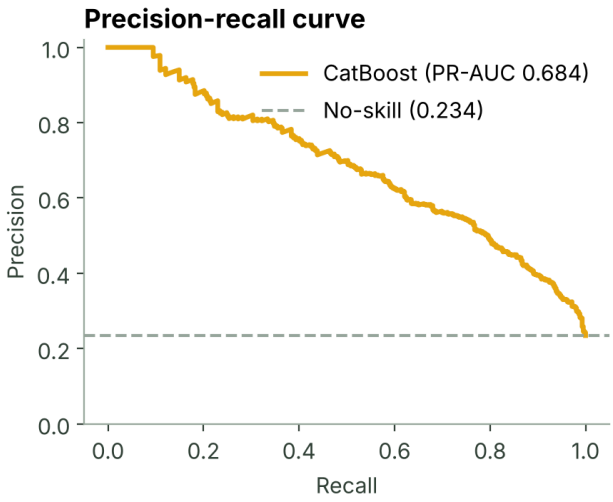
FINAL TEST RESULTS (1,800 UNSEEN CUSTOMERS)

Metric	Value
Accuracy	81.22%
Precision	59.46%
Recall	62.56%
F1 score	60.97%
ROC-AUC	0.8554
PR-AUC	0.6838



Of 422 churners in the test set, the model caught 264 and missed 158. It also raised 180 false alarms among 1,378 loyal customers.

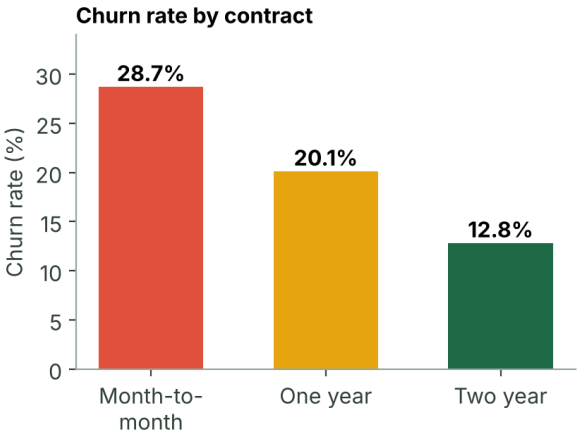




Left: ROC curve against a random baseline. Right: precision-recall curve against the no-skill line at the 23.4% churn rate.

Reading the recall. Test recall (62.6%) is a little below the 65% target set on validation data. A small gap like this is normal when moving from validation to unseen data, and it is reported here as it came out rather than tuned away.

What drives *churn*

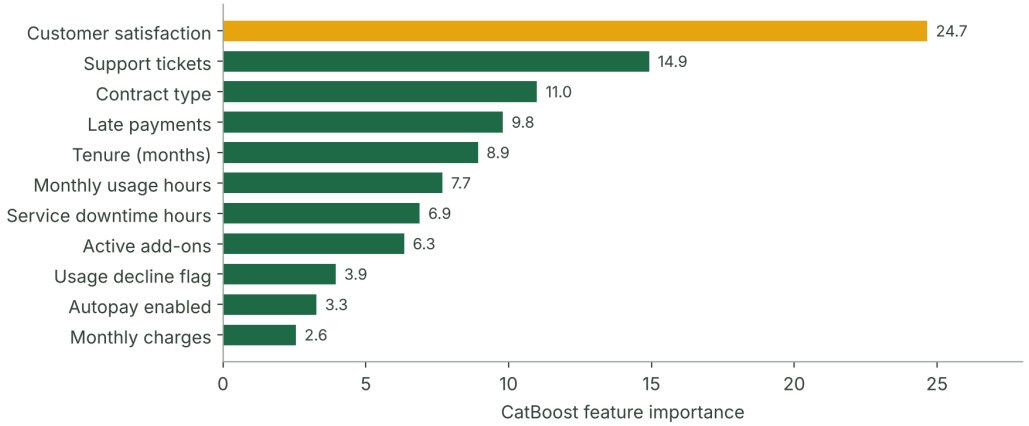


Observed churn rate by contract type, all 12,000 customers.

CHURNED VS RETAINED (AVERAGES)

Measure	Churned	Retained
Satisfaction	6.34	7.41
Support tickets	1.79	1.15
Late payments	0.99	0.62
Tenure (months)	32.1	37.6

TOP MODEL DRIVERS



CatBoost feature importance. Customer satisfaction and support tickets together account for roughly 40% of total importance.

READING THE PATTERNS

- **Unhappy customers leave.** Satisfaction is the strongest signal, followed by support tickets. A drop in satisfaction after a ticket is the earliest warning sign.
- **Contracts protect.** Month-to-month customers churn at 28.7%, more than twice the 12.8% on two-year contracts.
- **Payment friction matters.** Late payments rank fourth, which supports a billing-support action for these customers.

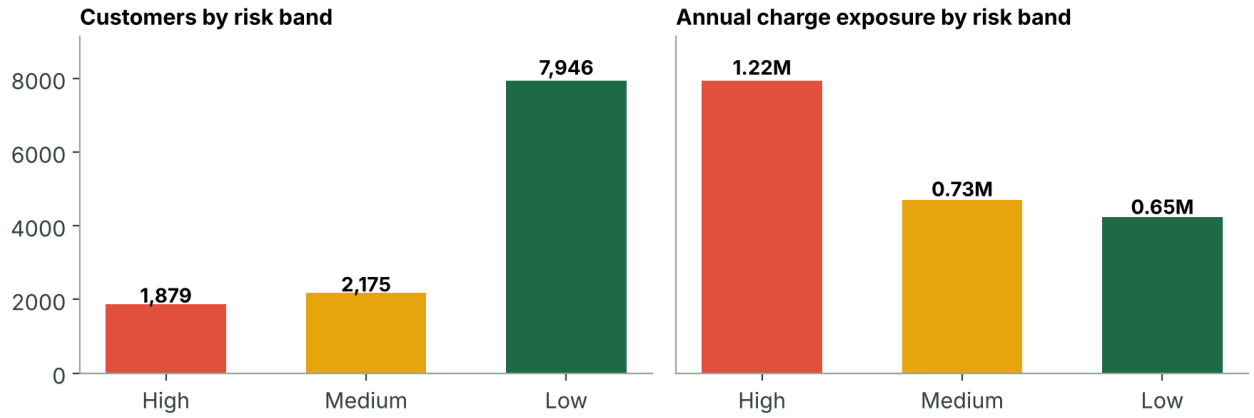
> **Caution.** These relationships were built into the simulation, so they show that the model recovers them, not new facts about real customers. Feature importance describes what the model uses to predict. It does not prove cause.

5 · FROM PREDICTION TO ACTION

Risk bands and *retention actions*

The model scored all 12,000 customers. Probabilities were grouped into bands, and each band maps to an action the retention team can take.

Risk band	Rule	Customers	Share	Charge exposure
High	Probability above 50%	1,879	15.7%	1,224,028
Medium	25% to 50%	2,175	18.1%	725,702
Low	Below 25%	7,946	66.2%	652,658
Total		**12,000**	**100%**	**2,602,388**



High-risk customers are 15.7% of the base but hold 47% of the annual charge exposure.

RECOMMENDED ACTIONS

Action	Who gets it	Customers
Priority service recovery call	High risk with 3 or more support tickets	674
Billing support and payment plan review	High risk with 2 or more late payments	404
Personalised retention offer	Other high-risk customers	801
Engagement and satisfaction check-in	Medium risk	2,175
Standard service; monitor	Low risk	7,946

Annual charge exposure is a prioritisation proxy (monthly charges × 12 × churn probability). It is not a forecast of recoverable revenue.

6 · LIMITATIONS AND NEXT STEPS

What this does *not* prove

LIMITATIONS

- **Simulated data.** Results hold only for this simulation and are not benchmarks for real customers.
- **Learnable by design.** The churn signals were built in, which makes the task easier than real data.
- **Random split.** Real deployments need time-based splits, since models predict the future from the past.
- **Importance is not causation.** Improving satisfaction may not reduce churn by the amount the model implies.
- **Exposure is a proxy.** It ranks customers but does not estimate the revenue a campaign would recover.

NEXT STEPS

- Repeat the experiment on a documented real dataset, such as IBM's public Telco Customer Churn sample.
- Set the threshold using campaign cost against customer value, not accuracy and recall alone.
- Add fairness checks and drift monitoring before any real use.
- Build a Power BI dashboard on the exported CSV for the retention team.

REPRODUCING THE PROJECT

Item	Detail
Notebook	Customer_Retention_Advanced.ipynb (fixed random seed 42)
Output	customer_retention_predictions.csv (12,000 rows, 19 columns)
Libraries	Python, pandas, NumPy, scikit-learn, CatBoost, matplotlib
Source code	github.com/kevinn7305/Smart-Customer-Churn-and-Retention-Intelligenece-System-using-Machine-Learning-